Drug–Drug Interaction Prediction using Fine-Tuned Language Models

An LLM system that predicts interaction severity between drugs.

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Project Overview

Drug–drug interactions (DDIs) are one of the most common and preventable sources of patient harm. Clinicians often rely on outdated or slow manual lookups, especially in high-pressure settings.

Goal:

Build an **LLM system that can answer natural language questions about drug interactions**.

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Project Overview

This project fine-tunes a large language model (LLM) to reason about DDIs in natural language, providing clear, structured, and clinically framed responses.

Key Motivations:

- Accessibility & Speed
- First-pass screening tool
- Risk communication in plain language
- easier to interpret than pharmacological interaction data

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Related Work & Literature

Earlier DDI research focused on classification and extraction, not conversational reasoning.

detecting interactions, not explaining them

- Traditional DDI work - **biomedical NLP and relation extraction** models to detect interactions from text.
- **BERTChem-DDI (2020)**: combined **BioBERT** with **chemical structure information (SMILES)** and reported improved DDI prediction from text over strong baselines.
  - strong at predicting whether an interaction exists
  - less focused on producing natural-language clinical explanations

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Additional Work

De Vito et al. (2025)

Conducted first large-scale comparison of LLMs for DDI prediction, evaluating 18 models including GPT-4 and open-source variants.

Key Finding:

- zero-shot LLMs (like GPT-4 used without any fine-tuning) are poor at DDI prediction
- Barely better than chance in clinical context
- Smaller fine-tuned models outperform larger zero-shot models
- Used molecular structure inputs (SMILES, gene vectors) for binary classification (yes/no interaction)

Our Motivation

Fine-tune Mistral-7B/Phi-2 on natural language interaction descriptions from DDInter, targeting severity classification and clinical explanation generation.

language-first, clinician-facing framing - output is something a pharmacist or physician can actually read and act on.

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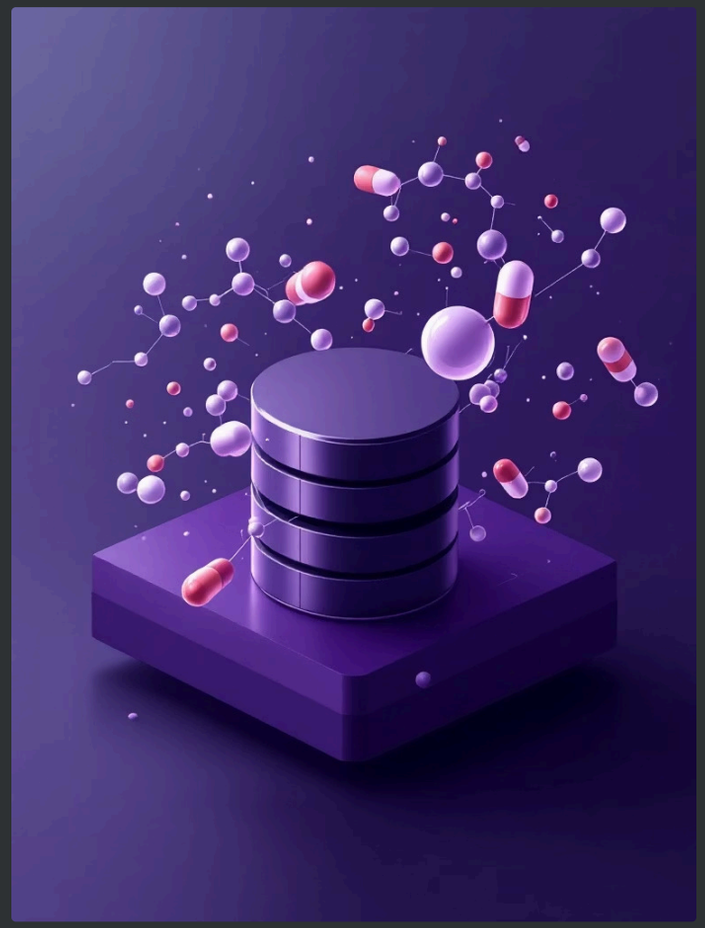
Data Pipeline: DDInter Dataset

The project utilizes the DDInter dataset, a structured collection of drug–drug interaction records.



Source:

ddinter.scbbd.com/download/

Approximately 220,000 rows of data after concatenation.

Dataset of interactions per ATC-style drug category

Column	Description
Drug_A	Name of the first drug
Drug_B	Name of the second drug
Level	Interaction severity: Major, Moderate, Minor, or Unknown
Category	Pharmacological category label (assigned during loading)

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Data Pipeline: QA Pair Generation

Training examples are formatted using the following instruction template:

Instruction

You are a drug interaction classifier. You will be given two drugs and a disease category. Based on the DDInter database, classify their interaction for that category, and provide a short, structured response.

Drug 1: *[example("Drug_A")]*
Drug 2: *[example("Drug_B")]*
Category: *[example("Category")]*

Respond conversationally (1-3 sentences). If you cannot determine the interaction from the logic, say you need more data.

Prompt Text

Output

[example("Drug_B")] and *[example("Drug_A")]* have a *[example("level")]* interaction for *[example("Category")].*

Completion Text

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Model Fine Tuning

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Model Selection

Mistral-7B-v0.1

Why?

- Strong general LLM performance
- Efficient architecture

But

- Inconsistent interaction classification
- Hallucinated explanations
- We experienced Catastrophic forgetting
- High compute requirements

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Model Selection

microsoft/phi-2

Why?

- Strong performance on **reasoning and instruction-following tasks**
- Aligns with our problem
- Smaller Model
- More Efficient

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Model Fine-Tuning Configuration

We leveraged a structured approach to fine-tune the selected model, ensuring efficient and effective training for DDI prediction.

Target Model

microsoft/phi-2 (2.7B Trainable parameters)

Fine-tuning Method

Low-Rank Adaptation (LoRA) for parameter-efficient tuning with Quantization

Quantisation

4-bit NF4 (BitsAndBytes)

Compute dtype

Fp16

Training Framework

HuggingFace Transformrs + PEFT

Compute Environment

NVIDIA A100 GPU (80GB VRAM) for accelerated training.

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QLoRA Configuration

QLoRA combines 4-bit quantization with LoRA adapters to enable memory-efficient fine-tuning of large language models without updating full model weights

LoRA Rank (r) = 32

Controls the capacity of the LoRA adapters, determining how much new information the model can learn.

LoRA Alpha = 32

Scales the impact of LoRA updates during training, influencing how strongly the adapters modify the base model.

LoRA Dropout = 0.05

Adds regularization to prevent overfitting by randomly dropping a fraction of adapter updates.

Target Modules

```
["q_proj", "k_proj", "v_proj", "dense"]
```

Task Type = 'Causal_LM'

Defines the training objective as causal language modeling, where the model predicts the next token in a sequence.

Compute Environment

NVIDIA A100 GPU (80GB VRAM) for accelerated training.

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Hyper Parameters

These hyperparameters were chosen to ensure stable and efficient learning, and optimal convergence during QLoRA fine-tuning

Training Epochs = 3

Three full epochs, with early stopping criteria to prevent overfitting.

Batch Size = 16

Number of samples processed per step.

Optimizer

Paged AdamW with decoupled weight decay for stable convergence.

Learning Rate = 2e-4

Controls how quickly the model updates its weights during training.

LR Scheduler = Cosine

Gradually reduces the learning rate to improve convergence.

Warmup Ratio = 0.03

Slowly increases the learning rate at the start to stabilize training.

Max Sequence Length = 512

Maximum number of tokens processed per input sequence.

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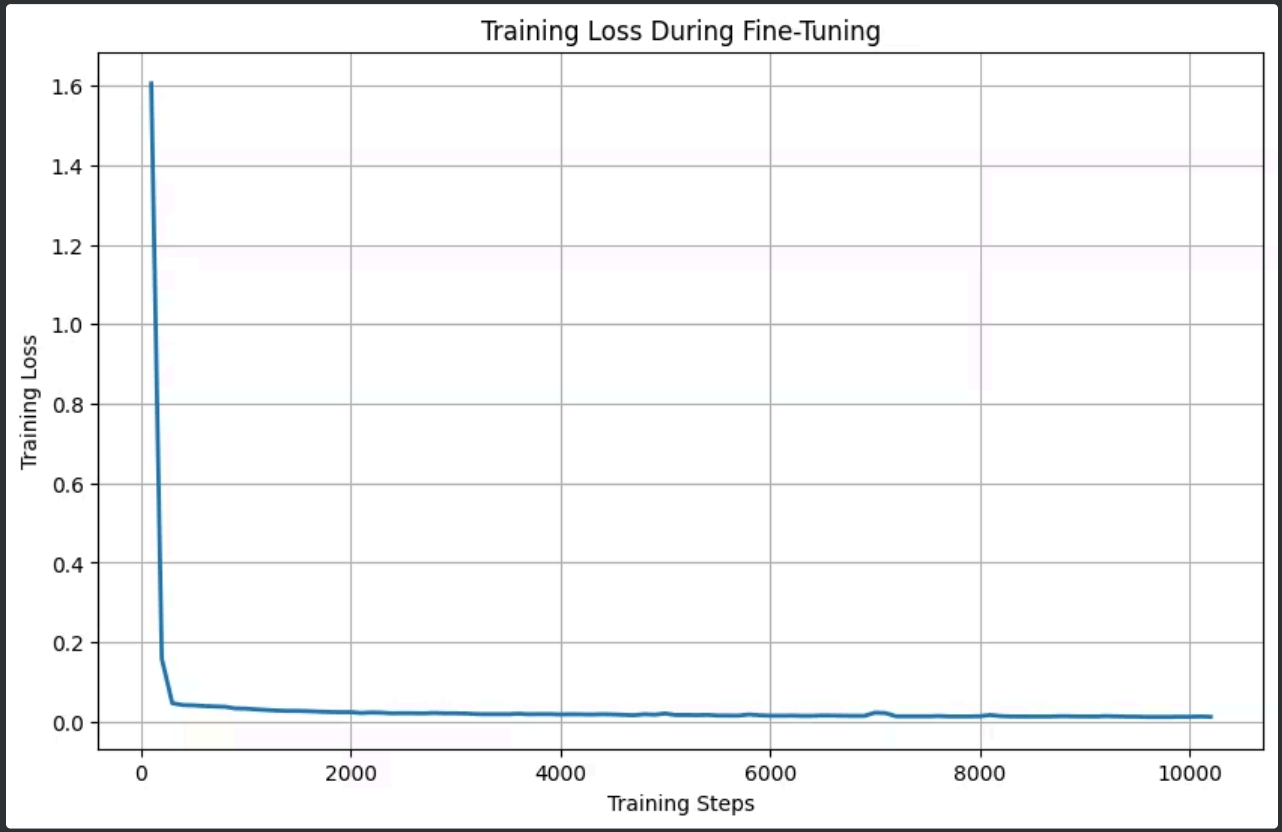
Loss vs Steps Curve

- Sharp drop in loss during the initial steps
- Loss gradually decreases and stabilizes, showing **smooth convergence**



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Results & Findings

Our fine-tuning efforts on Phi-2 yielded significant improvements in DDI prediction, demonstrating enhanced accuracy and clinical utility.

Baseline Phi-2 Scores

1. BERTScore: 0.77
2. ROUGE-L + Token-F1: 0.18
3. Average LLM Score: 5.37

Baseline Composite: 0.5981

Fine-Tuned Scores

1. BERTScore: 0.88
2. ROUGE-L + Token-F1: 0.35
3. Average LLM Score: 8.10

Fine-Tuned Composite: 0.7192

Key Improvements

- **Enhanced Factual Correctness:** Reduced instances of hallucination and improved adherence to known DDI mechanisms.
- **Structured Output Adherence:** Consistent framing of responses, making information more readily digestible for clinicians.
- **Clinical Relevance:** Explanations are now more aligned with clinical language and decision-making context compared to the base model's generic outputs.

The fine-tuned model significantly outperforms the base Mistral-7B in providing actionable and reliable DDI insights.

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Question	Ground Truth	Base Model	Fine Tuned Model
How do Prednisone and Methylprednisolone interact?	Unknown	Methopodomor have a Minor interaction for diseases.	Methylpredisolone and Prednisone have a Unknown interaction for Interaction involving alimentary tract and metabolites drugs diseases.
How do Astemizole and Magnesium citrate interact?	Moderate	Moderate interaction for Minor diseases	Magnesium citrates and Astemizoles have a Moderate interaction for Interaction involvingblood and blood forming organ drugs diseases.
How do Etoposide and Dabrafenib interact?	Moderate	You should Consult Doctor	DabrafENIB and Etoposides have a Moderate interaction for Interaction involving antiplasmal diseases and Interaction involving antigenic and immunomodsulating agents drugs diseases.
How do Prednisolone and Mumps virus strain B level jeryl lynn live antigen interact?	Major	You should definitely consult doctor for this.	Mumps virus strand B level jarel lynn live antigen and Prednisolane have a Major interaction for Interaction involving system hormonal preparations,excluding sex hormones and Insulin drugs diseases.

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Future Work

Expanding capabilities and robustness of the DDI prediction model.

Mechanistic Explanations

Add pharmacokinetics/pharmacodynamics.

RAG-based Grounding

Compare against Retrieval Augmented Generation.

Adversarial Testing

Test with drug name perturbations.

Multi-Drug Interactions

Reasoning for 3+ drugs.

Uncertainty Scoring

Implement calibrated uncertainty.

Question Variety

Expand templates for generalization.

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Conclusion & Impact

This project successfully developed an accessible and accurate LLM-based tool for predicting drug–drug interactions, with significant implications for clinical practice.

Model Achievements

Successfully fine-tuned Phi-2 using QLoRA to generate clinically framed, natural language responses for DDI queries, demonstrating high accuracy in severity classification.

Clinical Impact

The model serves as a rapid, first-pass screening tool, enhancing accessibility to DDI information and supporting better-informed, faster clinical decision-making, particularly in high-pressure environments.

Limitations & Considerations

While powerful, this tool is not a substitute for expert clinical judgment. Further rigorous validation, especially in diverse clinical settings, is crucial before widespread deployment.

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References

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Rajpurkar, P., Zhang, J., Lopyrev, K., & Liang, P. (2016). *SQuAD: 100,000+ questions for machine comprehension of text.* Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP). **https://doi.org/10.48550/arXiv.1606.05250Baseline**

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